



Brian Juma

Robotics & AI Engineer

Contact

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Github profile:

<https://github.com/Okelo123>

Expertise

Programming

Languages : Python,

JavaScript, C++,Html,

Css,

Bootstrap, React and

PHP.

IoT & Robotics

PROFESSIONAL SUMMARY

Robotics & AI Engineer skilled in building intelligent systems, IoT automation, and AI-driven solutions. Experienced in developing robotics prototypes, automation systems, health tech solutions, and smart devices showcased in innovation events and hackathons. Strong background in embedded systems, software engineering, cloud tools, and team leadership.

TECHNICAL SKILLS

Robotics & IoT: Arduino, ESP32, Sensors, Actuators, Embedded Systems, Automation

Programming: Python, C++, JavaScript, PHP, HTML/CSS, React

AI & Data Science: Machine Learning, EDA, Data Visualization, Model Evaluation

Tools: Git/GitHub, MySQL, Linux/Ubuntu, Basic AWS & Azure

Soft Skills: Problem-solving, Leadership, Documentation, Creativity

EXPERIENCE

- Robotics Engineer – SalamaFarm (Lead Hardware & Robotics Engineer) (2024 – Present)
- Served as the Lead Robotics & Hardware Engineer, coordinating the hardware team in developing advanced farm data acquisition sensors.
- Led the design and implementation of moisture and temperature sensors for the crop monitoring system, enabling accurate real-time environmental analysis.
- Played a major role in building an AI-powered farm monitoring system using ESP-CAM, moisture, and pH sensors for early crop disease detection and automated recommendations.
- Collaborated closely with the SalamaFarm hardware team to integrate robotics, IoT, and AI for precision agriculture solutions.
- Improved agricultural workflows by automating data collection, remote monitoring, and intelligent analysis.
- Developed automated software systems and AI-driven features.
- Provided consultancy on cloud, cybersecurity, and scalable software solutions.
- IoT & Tech Mentor – Community Programs at Zetech University (2024 – 2025)
- Conducted training on IoT, robotics, Git, and cloud technologies. such as AWS
- Led workshops, hackathons, and innovation sessions.
- IoT Lead – Zetech University (Research & Innovation Department) (2023 – 2024)
- Championed IoT and Robotics innovation within the university.
- Built flagship robotics and IoT prototypes: Robotic Smart Car, Smart Trash System, TrackBot Inspection Robot.
- Participated and represented the university in IoT competitions with Smart Traffic Light System and Smart Parking System.
- Awarded recognition as a Frontliner Innovator for outstanding performance. at Zetech university IoT hackathon
- Led multiple IoT learning sessions, improving student competency in IoT and robotics at Zetech University from 2024 to present.
- Supported Zetech University innovation events and guided students in project design and implementation. on projects and organizing events through ITCH community in Zetech and as IoT lead in Zetech

Innovation Week Robotic Projects

IoT Hackathon Projects

Community-Built IoT Systems (As Lead) at Zetech University

Bachelor of Science in Software Engineering (2023 – 2025)

Power Learn Africa Scholarship Program (PLP) (Feb 2025 – Nov 2025)

CERTIFICATIONS

IoT Community Lead at Zetech University

plp final project Group lead

LANGUAGES

PROJECTS

- Robot Smart Car (autonomous navigation)
- Smart Trash System (automated waste management)
- TrackBot Inspection Robot (terrain inspection)

- Smart Parking System
- Smart Traffic Light System

- Smart Security Light System
- Smart Smoke Detector
- Smart Distance Alert System
- Intruder Detection Alarm
- Smart Circuit Breaker
- Smart Door Lock System

EDUCATION

Zetech University | 2023 – 2026

Focused on: Web Technologies, Database Management, Startup Building, Employability Skills, Software Engineering, and AI for Software Engineering.

- AI in Software Engineering – PLP
- Software Development – PLP
- HCIA IoT – Huawei
- Circular Business Designer – Circular Innovation Hub

LEADERSHIP EXPERIENCE

- IoT Community Lead at Zetech university – Led hands-on IoT sessions and teams.
- Mentor in robotics, cloud, Data Science, and AI.
- Organized tech events, workshops, and hackathons at Zetech University.

- Led development of a heart-rate sensor system; ensured timely delivery and successful program graduation. at Power Learn Project Africa (plp).

- Focused on SDG 3: Good Health and Well-being.

English

Kiswahili

REFERENCES

Available upon request.